

Final Verification Report

Conversational Memory Intelligence System

Deliverable	6 - Implementation and Independent Verification
Author	Ketan Juikar
Verification date	2026-07-18
Scope	Milestones M0-M6 (implementation/), re-verified fresh for this deliverable

This report is the standalone verification/final_verification.pdf artifact required by the handbook's Deliverable 6. It consolidates two layers of verification: (1) the per-milestone L4 VERIFY passes performed by a separate model instance during the build, and (2) a fresh, independent re-run performed specifically for this deliverable against a newly created database volume, plus one live database-level security spot-check outside the application layer entirely. Raw evidence backing every claim below lives in verification/results/.

1. Scope and Methodology

The system under verification implements all seven Genesis loops defined in .genesis/PLAN.md (M0 foundation and tenant isolation, M1 write gate and secrets filter, M2 hybrid search and ranking, M3 contradiction handling, M4 forgetting and deletion, M5 observability and the Deliverable-3 benchmark re-run, M6 stretch prototypes). This report verifies the system as a whole against the handbook's Deliverable 6 minimum acceptance checks, not milestone by milestone.

What 'independent verification' means here

Per the handbook: verification must be performed from the specification and acceptance criteria, not by accepting the implementation agent's own assessment. Two layers were applied: every milestone was closed by a separate model instance (L4 VERIFY) given only the goal, success criteria, and invariants -- not the build's reasoning -- and instructed to independently re-derive evidence rather than trust the build's claims. For this deliverable specifically, the full test suite, static analysis, and benchmark were re-run again from a freshly recreated database volume, and one additional live database security check was run directly via psql, bypassing the application layer.

Commands re-run for this report (raw output in verification/results/)

- docker compose down -v && docker compose up -d (fresh volume, no accumulated state)
- pytest tests/ -v -> 71/71 passed
- mypy --config-file mypy.ini . -> clean, 44 files
- ruff check . -> clean
- python -m benchmark --compare-baseline experiments/baseline_results.csv
- psql -U cmis_app -d cmis (live RLS spot-check, independent of pytest)

2. Functional Verification Results

Test file	Tests	Proves
test_tenant_isolation.py	5	INV-1: zero cross-tenant leak; DB refuses unfiltered queries
test_write_gate.py	17	INV-2: secrets blocked (9 cases), innocent text not blocked (4)
test_retrieval.py	11	Hybrid search, relevance floor, token budget (INV-3), isolation
test_contradiction.py	10	C8: confidence-gated supersede, auditable history
test_forgetting.py	13	Reweight/archive correctness, INV-4 deletion window
test_observability.py	6	Health numbers, C13/INV-7 graceful 503 degradation
test_prototypes.py	7	M6 stretch: reranker + classifier prototypes, off by default

Total: 71/71 tests passed on a freshly recreated database volume. mypy and ruff both clean.

Deliverable 6 minimum acceptance checks (handbook Sec. 8.3)

Acceptance check	Result
Relevant memories outrank distractors	PASS - test_retrieval.py (2 tests)
Stale preferences do not override current	PASS - stale_wins_rate 0.333 -> 0.000
No cross-tenant memory under adversarial queries	PASS - live RLS check + 7 tests
Deletion removed from storage and retrieval paths	PASS - test_forgetting.py (4 tests)
Context selection respects token budget	PASS - test_never_exceeds_token_budget
Sensitive-data policy tested both directions	PASS - 9 positive + 4 negative cases
Benchmark compared against naive baseline	PASS - see Section 3

3. Benchmark vs. Naive Baseline (Deliverable 3)

Same seeded workload (seed=42) driven through the real write path. Fresh re-run against a clean volume; store size 63 memories at comparison.

Metric	Baseline	Real system	Gate
cross_tenant_leak_rate (S4)	0.250	0.000	Hard gate - MET
secrets_stored (S5)	2.000	0.000	Hard gate - MET
stale_wins_rate (S2)	0.333	0.000	Hard gate - MET
abstention_failure_rate (S6)	1.000	0.333	Improved, not zero
precision/recall/hit@5 (S1)	0.200/1.000/1.000	0.200/1.000/1.000	Matches exactly
retrieval p95 latency	0.69ms	18.73ms	Under 100ms budget

Residual gap, reported not concealed

S6 abstention_failure_rate improved (1.000 -> 0.333) but did not reach zero. PLAN.md's M2 success criteria treats this as a directional improvement, not a hard zero-gate like S2/S4/S5. The relevance floor is documented as a tunable starting guess; not re-tuned for this deliverable since it sits outside Deliverable 6's freeze boundary. Full detail: [verification/results/benchmark_comparison.md](#).

4. Security Verification

Full detail and evidence citations: [verification/security_report.md](#). Summary of the threats named in [design/threat_model.md](#), checked independently for this deliverable:

Threat	Status	Evidence
T1/T7 Cross-tenant read / spoofed identity	Holds	Live RLS spot-check + 7 tests
T2 Secrets stored	Holds*	17 write-gate tests, both directions
T3 Memory poisoning	Holds for tested cases*	10 contradiction tests
T5 Incomplete deletion	Holds	4 forgetting/deletion tests

* Residual risk explicitly named in [threat_model.md](#) and not claimed closed: novel secret formats can still slip past pattern-based detection; an AI-judge write gate can in principle be manipulated by adversarial text. Both are mitigated (encryption at rest, audit trail, a classifier-graduation path prototyped in M6) rather than eliminated.

Database role boundaries (verified from [db/schema.sql](#))

Role	Privileges	Reachable via HTTP?
cmis	Table owner, superuser	Never (schema setup only)
cmis_app	NOBYPASSRLS; scoped CRUD grants	Yes - the app's own role
cmis_job	BYPASSRLS; 3 narrow grants only	No - confirmed by inspection

5. Overall Verdict

All Deliverable 6 minimum acceptance checks pass on a freshly re-run, independently verified basis. Every hard-gate metric from the handbook and from this project's own PLAN.md (cross-tenant leak, secrets stored, stale-preference wins) reaches exactly zero. mypy and ruff are clean; 71/71 tests pass fresh.

Two things are reported as open rather than hidden, consistent with the handbook's standard that known failures must be surfaced, not concealed:

- S6 abstention_failure_rate improved from 1.000 to 0.333 but is not yet zero -- a directional-improvement target per PLAN.md, not one of the three hard-zero gates.
- The M6 write-gate classifier prototype's 1.000 held-out agreement was measured against a rule-based judge stand-in, not the real fuzzier AI judge -- evidence the training mechanism works, not evidence the real judge is this cleanly learnable.

Verdict: PASS against Deliverable 6's minimum acceptance checks.

This verdict covers the implementation and its independent verification specifically. It does not by itself certify Deliverables 1-5, 7, or 8 of the handbook, which cover problem reconstruction, research scanning, the productive-failure baseline, system design, the Genesis workflow, the engineering journal, and knowledge transfer respectively -- see the project's other deliverable folders for that evidence.